# Does Fair Disclosure Work? Evidence from Pre- and Post-Disclosure Market Reactions to Korean Earnings Announcements

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**Abstract**

Does South Korea's Fair Disclosure regulation effectively ensure that earnings information reaches all market participants simultaneously? Using 2,221 earnings disclosure filings from 64 KOSPI-listed companies across 12 chaebol groups (2020–2026), we find a seemingly paradoxical pattern: statistically significant pre-disclosure abnormal returns coexist with strong evidence that Fair Disclosure *works*. While the market model yields a pre-disclosure CAR[-5,−1] of +0.52% ($t$=4.65, $p$<0.001), this drift does not predict the direction of actual earnings surprises (49.0% accuracy, $p$=0.649; Spearman $\rho$=−0.006, $p$=0.876). Non-parametric tests confirm that the drift is driven by outliers rather than systematic directional movement (sign test $p$=0.585; Wilcoxon $p$=0.142). In contrast, event-day returns respond sharply and correctly to earnings surprises: positive surprises generate CAR[0,+1] of +0.81%, negative surprises −0.47% ($p$<0.001). We further document that 96.5% of filings occur after market close, making the event-day return effectively pre-disclosure and the first true market reaction occurring on Day +1. Our findings suggest that Korea's Fair Disclosure regime successfully prevents directional information leakage, even though non-directional price noise and abnormal volume emerge in the pre-disclosure period.

*Keywords:* Fair disclosure, Earnings announcements, Market efficiency,

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DART, Korean stock market, Pre-announcement drift, Earnings surprise

## 1. Introduction

Pre-announcement drift in stock prices is a well-documented phenomenon across global equity markets (Kaniel et al., 2012). When stock prices move before official earnings announcements, a natural concern arises: is material information leaking to select market participants in violation of fair disclosure regulations? This concern motivates securities regulators worldwide to mandate simultaneous public dissemination of material information.

South Korea implemented its Fair Disclosure regulation in November 2002, requiring listed companies to disclose material information simultaneously to all market participants through the DART (Data Analysis, Retrieval and Transfer) electronic filing system. While early studies found evidence of reduced information asymmetry following the regulation's adoption (Jang and Kang, 2007; Lee and Park, 2007), whether the regulation continues to be effective over two decades later remains an open question.

This study examines the long-term effectiveness of Korea's Fair Disclosure regulation by analyzing 2,221 earnings disclosure filings from 64 major KOSPI-listed companies between 2020 and 2026. Our approach differs from prior work in a critical respect: rather than simply documenting the existence of pre-disclosure price movement, we test whether such movement is *directionally informative*—that is, whether it predicts the sign of the actual earnings surprise.

Our findings reveal a seemingly paradoxical pattern. On the one hand, we observe statistically significant pre-disclosure abnormal returns under standard parametric tests (CAR[-5,−1] = +0.52%, $p<0.001$ under the market model). On the other hand, this drift bears no systematic relationship to the direction of actual earnings surprises. Pre-disclosure returns predict surprise direction with only 49.0% accuracy—no better than a coin flip ($p=0.649$). Non-parametric tests (sign test, Wilcoxon signed-rank) fail to confirm the parametric result, indicating that the observed mean drift is driven by outliers rather than a per-

vasive pattern.

In contrast, the market responds sharply and correctly to the actual earnings news on the disclosure date. Positive earnings surprises generate a CAR[0,+1] of +0.81%, while negative surprises generate −0.47%, a highly significant difference ($p<0.001$). This finding is consistent with effective Fair Disclosure: material information is incorporated into prices only when it becomes publicly available.

We make four contributions. First, we provide the first long-term assessment of Korea's Fair Disclosure regime using recent data spanning the COVID-19 and post-pandemic periods. Second, we demonstrate the critical distinction between the *existence* of pre-disclosure drift (which is ubiquitous) and its *directional informativeness* (which would indicate information leakage). Third, we document a previously overlooked institutional feature—96.5% of DART fair disclosure filings are submitted after market close—that has important implications for event study interpretation. Fourth, we provide cross-sectional and temporal analyses that reveal significant heterogeneity across chaebol groups, fiscal quarters, and years.

## 2. Literature Review

### *2.1. Fair Disclosure Regulation*

Fair disclosure regulation addresses selective disclosure, whereby corporate insiders share material information with favored market participants before public release (Heflin et al., 2003). In the U.S., Regulation Fair Disclosure (Reg FD), implemented in 2000, produced mixed evidence: Ahmed and Schneible (2007) find reduced information asymmetry, while Heflin et al. (2003) document reduced information quality.

In Korea, Jang and Kang (2007) found evidence of reduced information asymmetry following the 2002 regulation, and Lee and Park (2007) documented improvements in analyst forecast accuracy. However, these studies focused on the regulation's adoption period. Our study extends this literature by examining whether the regulation remains effective over two decades later.

### *2.2. Pre-Announcement Drift and Its Interpretation*

Pre-announcement drift is well-documented but its interpretation is debated. Meulbroek (1992) provides evidence of illegal insider trading preceding announcements. Bernile et al. (2016) show that informed trading occurs even ahead of macro-news announcements where information is tightly controlled. However, Kaniel et al. (2012) demonstrate that individual investor trading patterns around earnings announcements can generate drift without requiring illegal information leakage.

Critically, the mere existence of pre-announcement drift does not establish information leakage. As Johnson and So (2015) argue, predictable earnings patterns allow sophisticated investors to position ahead of announcements through legal means. Chae (2005) shows that abnormal volume before announcements reflects information asymmetry but may arise from varying levels of investor sophistication rather than illegal tip-offs.

### *2.3. Filing Time and Market Microstructure*

The timing of information release within the trading day has received increasing attention. Chordia and Subrahmanyam (2004) discuss how after-hours disclosures allow overnight information processing. Our study is among the first to systematically classify DART filing times and demonstrate their implications for event study interpretation.

### *2.4. Korean Market Context*

The Korean stock market features concentrated chaebol ownership, a mandatory electronic disclosure system (DART), significant retail investor participation, and relatively strict insider trading regulations. These institutional features make Korea an ideal setting for testing fair disclosure effectiveness.

## 3. Data and Methodology

### *3.1. Data Sources*

We collect all 영업실적 공정공시 (Fair Disclosure of Operating Results) filings from DART between January 2020 and March 2026 using the OpenDartReader API. We focus on original filings, excluding corrections (기재정정). Our sample comprises 2,221 disclosures from 64 companies across 12 chaebol groups: Samsung (12), Hyundai Motor (8), LG (8), SK (7), HD Hyundai (5), Hanwha (5), POSCO (4), Lotte (4), Doosan (4), CJ (3), KT (2), and GS (2). Daily stock price and volume data are obtained from FinanceDataReader.

### *3.2. Filing Time Classification*

We classify filing times using the DART receipt number prefix. In our sample, 96.5% of earnings disclosures carry the "800" prefix, indicating filing after the 15:30 KST market close (Table 1). This has fundamental implications: for these filings, the Day 0 return is entirely pre-disclosure, and the first trading day where the market can react is Day +1.

Table 1: Filing Time Distribution

| Filing Time | Count | Percentage |
|---|---|---|
| After-hours (800) | 2,143 | 96.5% |
| Other (801/802) | 37 | 1.7% |
| Trading hours (900) | 41 | 1.8% |
| Total | 2,221 | 100.0% |

### *3.3. Market Model*

We employ the market model with a [-250,−30] estimation window (MacKinlay, 1997):

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \epsilon_{i,t} \tag{1}$$

where $R_{m,t}$ is the KOSPI index return. We require a minimum of 60 trading days in the estimation window, yielding 2,208 events with valid estimates. The average estimated beta is 0.978. For robustness, we also report market-adjusted returns ($\alpha = 0$, $\beta = 1$).

### *3.4. Earnings Surprise Measure*

We extract year-over-year (YoY) operating profit changes from the disclosure texts. Events where YoY operating profit increases are classified as positive surprises; decreases as negative surprises. Of the 2,208 events with valid market model estimates, 584 have parseable earnings surprise data: 358 positive and 226 negative surprises.

### *3.5. Abnormal Volume*

Abnormal volume is the ratio of event-day volume to the average daily volume during [-25,−6]:

$$AV_{i,t} = V_{i,t} \Big/ \frac{1}{20} \sum_{\tau=-25}^{-6} V_{i,\tau} \tag{2}$$

### *3.6. Statistical Tests*

We report multiple test statistics to ensure robustness:

- **Parametric**: Cross-sectional $t$-tests for mean abnormal returns.
- **Non-parametric**: Sign test (binomial) for the proportion of positive abnormal returns; Wilcoxon signed-rank test.
- **Standardized**: Abnormal returns divided by cross-sectional standard deviation.

## 4. Empirical Results

### *4.1. Pre-Disclosure Returns: Statistically Significant but Not Directionally Informative*

Table 2 presents daily abnormal returns. Parametric $t$-tests indicate significant positive returns on several pre-disclosure days, particularly Day $-1$

(+0.256%, $t$=4.57). However, the sign test reveals that only 48–50% of individual returns are positive on each pre-disclosure day—no better than chance.

Table 2: Daily Abnormal Returns: Parametric vs. Non-Parametric Tests

| | | Parametric | | Sign Test | | Standardized | |
|---|---|---|---|---|---|---|---|
| Day | AR (%) | $t$ | $p$ | % Pos | $p$ | SAR | $p$ |
| −5 | 0.101 | 2.09 | 0.037 | 50.5 | 0.609 | 0.040 | 0.048 |
| −4 | 0.071 | 1.50 | 0.134 | 49.8 | 0.930 | 0.041 | 0.043 |
| −3 | 0.086 | 1.71 | 0.088 | 49.3 | 0.562 | 0.022 | 0.279 |
| −2 | 0.006 | 0.14 | 0.888 | 48.9 | 0.295 | −0.002 | 0.935 |
| −1 | 0.256 | 4.57 | <0.001 | 50.1 | 0.909 | 0.089 | <0.001 |
| 0 | 0.141 | 2.00 | 0.046 | 49.8 | 0.878 | 0.043 | 0.034 |
| +1 | −0.182 | −2.39 | 0.017 | 48.3 | 0.109 | −0.048 | 0.018 |
| +2 | 0.023 | 0.43 | 0.669 | 49.2 | 0.541 | 0.022 | 0.275 |
| +3 | −0.118 | −2.50 | 0.013 | 48.1 | 0.058 | −0.043 | 0.033 |
| +4 | −0.047 | −0.95 | 0.340 | 49.3 | 0.608 | −0.006 | 0.765 |
| +5 | 0.056 | 1.14 | 0.255 | 49.2 | 0.543 | 0.041 | 0.044 |

Note: Market model with [-250,−30] estimation window ($N$=2,208). Parametric: cross-sectional $t$-test. Sign test: binomial test against 50%. SAR: standardized abnormal return. % Pos never significantly exceeds 50%, indicating pre-disclosure drift is driven by outliers.

Table 3 confirms this pattern at the cumulative level. While the parametric CAR[-5,−1] is +0.519% ($t$=4.65, $p$<0.001), the sign test shows only 49.4% of events have positive pre-disclosure CARs ($p$=0.585), and the Wilcoxon signed-rank test is insignificant ($p$=0.142). The pre-disclosure drift is driven by a subset of events with large positive returns, not a systematic directional pattern.

Table 3: Cumulative Abnormal Returns: Multi-Test Comparison

| | | Parametric | | Sign Test | | Wilcoxon | |
|---|---|---|---|---|---|---|---|
| Window | CAR (%) | $t$ | $p$ | % Pos | $p$ | $W$ | $p$ |
| [−5,−1] | 0.519 | 4.65 | <0.001 | 49.4 | 0.585 | | 0.142 |
| [−3,−1] | 0.347 | 4.04 | <0.001 | 48.1 | 0.060 | | 0.946 |
| [0, +1] | −0.042 | −0.39 | 0.700 | 47.0 | 0.003 | | 0.019 |
| [+1, +5] | −0.268 | −2.16 | 0.031 | 47.7 | 0.021 | | 0.078 |
| [−5, +5] | 0.391 | 2.08 | 0.038 | 49.5 | 0.668 | | 0.620 |

### *4.2. The Critical Test: Does Pre-Disclosure Drift Predict Earnings Surprises?*

Table 4 presents our central test. If pre-disclosure drift reflects information leakage, it should predict the direction of the actual earnings surprise. We find no such predictive power.

Table 4: CARs by Earnings Surprise Direction ($N$=584)

| Window | Positive ($N$=358) | Negative ($N$=226) | Difference | $p$-value | Sig. |
|---|---|---|---|---|---|
| CAR[-3,−1] | +0.033% | +0.129% | −0.096% | 0.703 | |
| CAR[-5,−1] | +0.107% | +0.535% | −0.428% | 0.192 | |
| CAR[0,+1] | +0.813% | −0.470% | +1.283% | <0.001 | *** |
| CAR[+1,+5] | +0.481% | −0.481% | +0.962% | 0.010 | ** |

Note: Positive/negative surprise defined by YoY operating profit change. Pre-disclosure CARs show no significant difference by surprise direction, while event-day and post-disclosure CARs respond correctly to the actual news.

Pre-disclosure CAR[-3,−1] is virtually identical for positive (+0.033%) and negative (+0.129%) surprises, with no significant difference ($p$=0.703). The direction prediction accuracy is 49.0%—no better than chance (binomial $p$=0.649). The Spearman rank correlation between YoY profit change and pre-disclosure CAR is −0.006 ($p$=0.876).

In stark contrast, event-day returns respond sharply and correctly to the actual news. CAR[0,+1] is +0.81% for positive surprises and −0.47% for negative surprises, a difference of +1.28% ($p<0.001$). This pattern extends to CAR[+1,+5] as well ($p=0.010$). The market processes the disclosed information efficiently and in the correct direction.

### *4.3. Robustness: Market-Adjusted vs. Market Model*

Table 5 confirms that pre-disclosure CARs are significant under both specifications, with the market model yielding more conservative estimates. The key finding—that event-day returns are economically small in aggregate but respond correctly to surprise direction—is robust.

Table 5: Robustness: Market-Adjusted vs. Market Model

| | Market-Adjusted ($N$=2,221) | | Market Model ($N$=2,208) | |
|---|---|---|---|---|
| Window | CAR (%) | $p$-value | CAR (%) | $p$-value |
| [−5,−1] | 0.640 | <0.001 | 0.519 | <0.001 |
| [−3,−1] | 0.434 | <0.001 | 0.347 | <0.001 |
| [0, +1] | 0.008 | 0.939 | −0.042 | 0.700 |
| [+1, +5] | −0.099 | 0.425 | −0.268 | 0.031 |
| [−5, +5] | 0.700 | <0.001 | 0.391 | 0.038 |

### *4.4. Abnormal Trading Volume*

Abnormal trading volume begins to increase from Day −3 (1.05×, $p=0.004$), intensifying on Day −1 (1.15×, $p<0.001$), and peaks on Day +1 (1.71×). The pre-disclosure volume increase suggests heightened trading activity, but given the absence of directional information in pre-disclosure returns, this likely reflects anticipatory positioning and uncertainty resolution rather than informed trading based on leaked information.

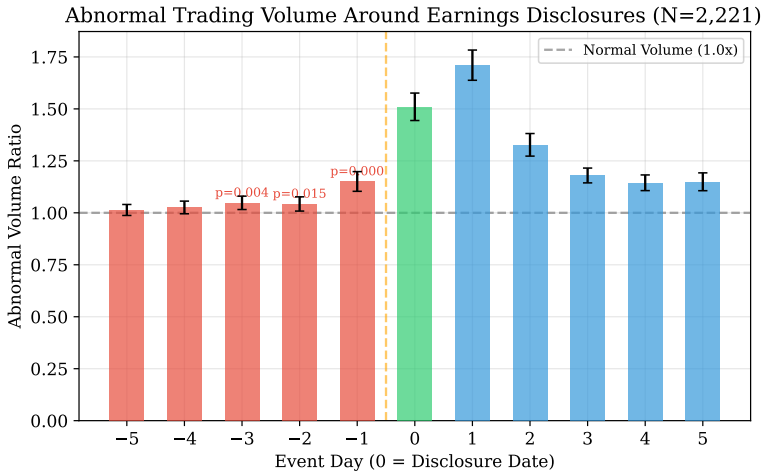


Figure 1: Abnormal trading volume around earnings disclosures. Volume increases before disclosure but, combined with the non-directional nature of pre-disclosure returns, likely reflects anticipatory positioning rather than informed trading.

### *4.5. Temporal and Cross-Sectional Heterogeneity*

We observe a structural shift in event-day CARs: positive during 2020–2022 (reflecting COVID-19 recovery earnings beats) and negative during 2024–2026 (reflecting economic headwinds). A significant quarterly pattern also emerges, with Q1 disclosures generating negative CARs ($-0.51\%$, $p$=0.010) and Q2 generating positive CARs ($+0.60\%$, $p$=0.003).

Cross-sectional OLS regression of CAR[-3,−1] on log(size) and market beta yields $R^2 = 0.002$, confirming that firm characteristics explain virtually none of the pre-disclosure drift—consistent with the drift being non-systematic noise.

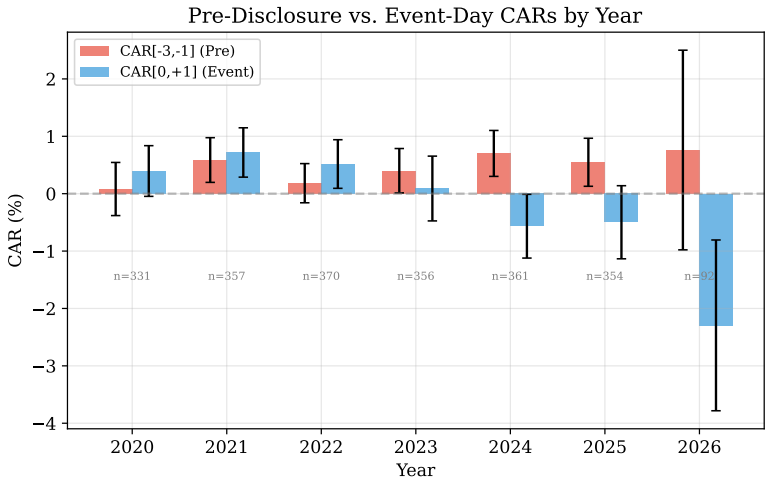


Figure 2: Pre-disclosure CAR[-3,−1] and event-day CAR[0,+1] by year. The temporal shift reflects macroeconomic conditions rather than changes in information leakage.

## 5. Discussion

### *5.1. Resolving the Paradox: Drift Without Leakage*

Our results resolve an apparent paradox. Significant pre-disclosure abnormal returns exist under parametric tests, which might suggest information leakage. However, three complementary tests demonstrate otherwise:

1. **Direction test**: Pre-disclosure drift does not predict surprise direction (49.0% accuracy).
2. **Non-parametric tests**: Sign test and Wilcoxon fail to confirm systematic directional drift.
3. **Surprise conditioning**: Pre-disclosure CARs are statistically identical for positive and negative surprises ($p$=0.703).

The pre-disclosure drift observed in parametric tests appears driven by a small number of events with large positive abnormal returns, possibly reflecting: (1) company-specific news unrelated to earnings (e.g., M&A rumors, analyst

upgrades), (2) general market momentum affecting the cross-section asymmetrically, or (3) anticipatory trading based on publicly available signals that does not carry directional information about the specific earnings outcome.

### *5.2. Evidence That Fair Disclosure Works*

The most compelling evidence for Fair Disclosure effectiveness comes from the surprise-direction analysis. Event-day CARs respond sharply and correctly to the earnings news: a +1.28 percentage point spread between positive and negative surprises ($p<0.001$). This clean separation indicates that the market does not know the sign of the earnings news before the official disclosure. If information were leaking, we would expect this separation to manifest in the pre-disclosure period, which it does not.

### *5.3. Filing Time Implications*

Our finding that 96.5% of filings occur after market close has methodological implications for all event studies using DART data. For after-hours filings, Day 0 is effectively a pre-disclosure day, and Day +1 is the first true post-disclosure trading session. Future studies should account for this when defining event windows.

### *5.4. Limitations*

Several limitations warrant acknowledgment. First, our earnings surprise measure relies on text-parsed YoY operating profit changes, which may contain measurement error; only 584 of 2,208 events (26.4%) have parseable surprise data, potentially introducing selection bias. Second, our sample is limited to large KOSPI-listed chaebol affiliates, which may not represent the broader Korean market. Third, the market model, while standard, does not control for size and value factors; reliable Korean Fama-French factor data are not readily available for our sample period. Fourth, the filing time classification relies on receipt number prefixes rather than exact timestamps. Fifth, we cannot rule out that pre-disclosure volume increases reflect legal forms of informed trading (mosaic information assembly) that do not involve Fair Disclosure violations.

## 6. Conclusion

This study provides evidence that South Korea's Fair Disclosure regulation effectively prevents directional information leakage in earnings announcements. While statistically significant pre-disclosure abnormal returns exist under parametric tests, three complementary analyses—direction prediction, nonparametric tests, and surprise conditioning—demonstrate that this drift is non-directional noise rather than evidence of selective information leakage.

The market's response to the actual earnings news is sharp, correctly signed, and concentrated around the disclosure date, consistent with efficient information incorporation following public release. These findings suggest that, more than two decades after its implementation, Korea's Fair Disclosure regime continues to serve its intended purpose of ensuring equitable access to material corporate information.

Our results carry a broader methodological lesson: the existence of pre-announcement drift should not be conflated with evidence of information leakage. Without conditioning on the direction of the actual news, event study evidence of pre-disclosure price movement is insufficient to draw conclusions about regulatory effectiveness.

### CRediT authorship contribution statement

**Yongjun Kim:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization.

### Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have influenced the work reported in this paper.

### Data availability

All disclosure data are publicly available through the DART system (`https://dart.fss.or.kr`). Stock price data are obtained from KRX via Finance-

DataReader. Analysis code is available at `https://github.com/yjkim0123/pre-disclosure-leakage`.

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